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AI-Enhanced Blockchain-Based Certificate Verification using Hybrid Deep Learning Models

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***Abstract*—In Advancement of digital world, the increase in the pace of issuing the digital certificates has also escalated the threat of fake educational qualifications, including in blockchain-based systems of verification, where the immutability cannot be enough to eliminate the registration of forged documents. In order to overcome this drawback, this research suggests an AI-enhanced blockchain-based certificate verification architecture that combines a hybrid deep learning model and the ElimuChain architecture. The proposed structure uses CNN-BiLSTM model that is able to identify visual and semantic inconsistencies in the certificates before anchoring them to the blockchain thus avoiding the possibility of forged certificates being stored permanently. The checked certificates are then hashed, hosted on the IPFS and permanently recorded on the blockchain through the smart contracts. Experimental assessment on a combined dataset of true and fake certificates proves that the proposed model has an accuracy of 98.3, precision of 97.9, recall of 98.6 and an F1-score of 98.2, which is better than standalone CNN and Transformer-based baselines. Trust, security, and reliability of nationwide systems of certificate verification are greatly increased based on the integration of AI-driven pre-validation.**

Keywords*—Blockchain, Educational Certificate Verification, Hybrid Deep Learning, CNN–BiLSTM, Certificate Forgery Detection, Smart Contracts, IPFS, Artificial Intelligence, Decentralized Applications, Trust Management.

I. INTRODUCTION

Educational certificates are important in verification of academic performance and professionalism of individuals. As the dependence on credentials to obtain a job, higher education, and a professional license grows, the integrity of certificates is turning into a burning issue on the global level. Even after the utilization of digital record management software, the issue of certificate forging has been ongoing, which has been facilitated by the sophisticated image editing software and the use of advanced document manipulation software [1], [2]. According to the recent research, the forged academic credentials still jeopardize the trust in the education systems, the labor markets, and the institutions, especially the cross-border ones [3]. The idea of using blockchain technology as a way of issuing and verifying certificates has gained a possibility of being a viable solution to the problem because of its decentralization, immutability, and resilience to tampering features [4], [5]. Certificates systems based on blockchain allow the verifiers to verify credentials without the help of centralized authorities, thus minimizing fraud, delays in verification, and administrative overheads [6]. Nevertheless, the majority of solutions that currently exist in the field of blockchain mainly assure post-issuance integrity with the understanding that the certificates posted to the blockchain are automatically authentic [7].

In recent works, an essential weakness of blockchain-based certificate verification systems has been pointed out: blockchain is not able to identify forged or manipulated certificates prior to registration [8], [9]. When a forged certificate has been hashed and stored on-chain, it becomes immutable, and thus makes fraudulent data permanent. The resulting issue is the so-called garbage-in, garbage-out issue, which is a massive constraint of the efficiency of blockchain-based verification systems in practical applications [10]. Concurrently, artificial intelligence and especially deep learning have shown high performance in forgery of documents through the analysis of visual features, text structure and semantic anomalies [11], [12]. Hybrid deep learning models Hybrid deep learning models combining convolutional neural networks (CNNs) with sequential learners have been demonstrated to be more effective in document authentication tasks [13]. Nevertheless, currently used AI-based systems to detect forgery are autonomous and do not have trusted, auditable, and decentralized storage solutions. Although this has been improved recently, there are still three major challenges remaining: Unidirectional non-hacks Lack of pre-issuance forgery detection in blockchain-based certificate systems. Non-existence of smart systems to ensure that forged certificates are anchored on-chain permanently. Weakly coupled AI-based trust validation and decentralized and nationalized certificate validation systems [14], [15]. These dilemmas lower the reliability of the systems and limit mass usage of blockchain-based educational credential validation.

This research will solve these limitations by presenting a proposal of an AI-enhanced blockchain certificate verification system that expands Elim Chain system with a hybrid CNN-BiLSTM deep learning model to detect forged certificates before registration in the blockchain. The suggested framework will implement a smart pre-validation layer, which scans the certificate images to identify the visual and semantic anomalies prior to the creation of cryptographic hashes to store in IPFS and anchor them in blockchains. The key contributions of this study can be summarized as follows: An AI-enabled certificate verification system that can work with hybrid deep learning and blockchain-based credential management on a seamless basis. CNN-BiLSTM forgery detecting model with a high success rate in detecting manipulated certificates prior to the on-chain storage. A

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secure and scalable verification pipeline, which includes AI validation, IPFS-based off-chain storage, and blockchain immutability). Much experimental analysis has shown increased detection precision and higher reliability than blockchain-only and standalone AI-based designs. The proposed system will greatly improve the reliability, security, and feasibility of large-scale educational certificates verification due to artificial intelligence combined with decentralized trust infrastructure [22].

II. RELATED WORK

A. *Certificate issuance and verification with the help of blockchain.*

It has been suggested extensively that blockchain can be used to offer tamper-resistant anchoring of the provided credentials, as well as, to streamline the third-party verification. They can be systems on university levels or national systems; a majority of the stores store certificate fingerprints (hashes) on-chain and full documents off-chain (e.g., IPFS) to minimize costs and enhance scalability. ElimuChain is an example of a nation-level design, which includes RBAC, government/regulatory enrollment, on-chain revocation and IPFS storage to facilitate multi-level education ecosystems [16].More privacy-sensitive versions have also been designed recently (such as ZK-enabled academic verification systems) to prevent the disclosure of sensitive properties and still enable verification to be checked [17, 24].

B. *AI-based detection of document / certificate forgery.*

Individually, deep learning models have been shown to be useful at detecting manipulated document images by training to identify small differences in textures, fonts, signatures and layout. More efficient cues (multimodal forgery cues) (CNN + sequential modules (BiLSTM or Transformer layers)) are presented by classic CNN architectures and enhanced hybrids. Hybrid CNN BiLSTM systems have also demonstrated state-of-the-art performance on adjacent detection problems (e.g., fake-news, document authentication), and exhibit greater resilience to varied kinds of manipulations and also layout changes [18]. Recently applied literature on the subject of academic document forgery detection boasts of a high accuracy rate with the utilization of domain-specific pipelines that integrate visual processing and low-resource hashing/checking steps [19].

C. *AI and blockchain integration (new trend).*

It is increasingly being understood that blockchain is insufficient to ensure that forged documents are anchored in a manner that is impossible to alter, and therefore, solutions to the problem propose the introduction of pre-issuance AI validation to eliminate garbage-in/garbage-out issues. Early integrated systems have already shown viability (AI pre-validation + blockchain anchoring), and other directions have been applying blockchain with verifiable ML primitives (e.g. ZK proofs of model inference or models outputs) to bring auditability and privacy to ML-driven checks [20, 23].

D. *Privacy, scalability and distributed model training.*

Credential systems Privacy-preserving protocols (i.e, zero-knowledge proofs (ZKP), federated learning (FL)) are becoming more commonly adopted: ZKPs can be used, without revealing the content, to verify that a piece of data is genuine, and FL can be used to train a common model across institutions without a central repository of sensitive data. Recent prototypes and surveys indicate that privacy leaks can be addressed by ZKP and layer-2 integrations and that FL can be applied to distributed fraud detection systems to learn cross-institution forgery signatures without releasing raw certificates [21,22, 25].

E. *Gaps in research*

Pre-issuance forgery detection is uncommon to blockchain processes.

The vast majority of blockchain certificate systems (even many operational prototypes) presuppose that uploaded documents are authentic and only provide immutability after they have been issued. This introduces a necessary weakness: when a forgery is hashed and stored in blockchain, the forgery will be forever (garbage-in immutable garbage). ElimuChain captures the infrastructure of systems to use in national implementation but fails to integrate AI validated in advance with the issuance pipe.

Hybrid visual models plus sequential are not well exploited in certificate authentication. Pure-CNN detectors: Visual artifacts Sequence models: Text/layout semantics Hybrid systems (e.g. CNN BiLSTM) may concurrently capture spatial representations and sequential textual/layout representations, however, little published research can be found to apply such hybrids specifically to the detection of educational certificate forgery on scale. The existing forgery detection articles addressed single institutions or neighboring areas (e.g. academic documents or fake-news), which means that hybrids can be customized and thoroughly tested on multi-format certificate archives. Here, there is a deficiency of verifiable AI + blockchain pipelines that do not infringe privacy.

New initiatives are investigating using ZK to validate the output of ML or construct ZK-enabled verification systems, but the connection between certificate issuance processes and ZK is early. Systems that can generate verifiable proofs indicative of an ML check passing (without leaking certificate contents) prior to on-chain anchoring would be much more trusted and simultaneously remain privately oriented, a combination that is difficult to achieve with the literature. The field of cross-institutional learning without the exchange of raw data has not been fully investigated in detecting forgeries. Patterns of forgery may differ among institutions, nations and templates of certificates. Privacy/regulatory A centralized ML training is frequently not possible. Principally, federated learning solutions to distributed forgery detection have not been thoroughly tested, where certificate datasets are heterogeneous, class imbalance, adversarial manipulation, and institutional drift are realistic.

There are operational issues (latency, compute, regulatory workflow) that are open. National roll-outs need to strike the right balance between the compute and latency added by AI checks and user experience and cost (e.g. mobile/edge vs. cloud inference). Published assessment of the combination of end-to-end metrics, namely, detection effectiveness, false positive effect on honest issuance, latency of verification, on-chain savings of costs by avoiding fraudulent anchoring, and governance procedures of disputed cases, is limited.

Table 1 Comparison of Related Certificate Verification Systems.

System	Multi-Level Support	RBAC	On-Chain Revocation	AI Pre-Validation	ZK Privacy / FL

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Gräther *et al.* (2022)	✗	✗	✗	✗	✗
Xie *et al.* (2023)	✗	✗	✓	✗	✗
Abreu *et al.* (2023)	✓	✓	✗	✗	✗
ZKBAR-V (2025)	✗	✗	✗	✗	✓
DL Forgery Detection (2023)	✗	✗	✗	✓	✗
ElimuChain (2025)	✓	✓	✓	✗	✗
Proposed	✓	✓	✓	✓	✓

Recent blockchain-based certificate verification systems have notably enhanced trust and transparency through decentralization and irreversibility; still, their functionality is still minimal, as it is summarized in Table 1. Early models are mainly based on credentials anchored on blockchains without mention of governance or revocation. Later researchers added role-based access control and on-chain revocation to improve institutional accountability, however, they are limited to post-issuance integrity checks. Privacy-concerned schemes that work with zero-knowledge proofs provide selective disclosure of credentials, but do not provide smart validation. Simultaneously, deep learning-based forgery detection models show a high level of performance in forging certificates, but they are not combined with blockchain infrastructures, which means that they are used in isolation. ElimuChain system goes a step further to offer the state of the art by enabling management at the multi-level certificate management with RBAC and on-chain revocation [16]. However, it still fails to stop forged certificates being registered forever. Table 1 has indicated that the proposed AI-enhanced system has the unique characteristics of combining hybrid deep learning-based pre-validation with blockchain verification, privacy preservation, and collaborative learning, which removes the essential gaps in the current response.

III. METHODOLOGY

Fig. 1 shows the architecture of the proposed framework, which is a continuation of the ElimuChain infrastructure with the addition of an intelligent pre-validation layer using hybrid deep learning. The design is made in such a way that authentic certificates are permanently imprinted on the blockchain to overcome the inherent weakness of the blockchain systems to identify forged inputs before registration. The system is composed of five key elements, namely (i) authorized certificate issuers, (ii) AI-based forgery detector, (iii) decentralized off-chain storage based on IPFS, (iv) blockchain network with smart contracts, and (v) certificate verifiers. These elements are communicating with each other by means of a decentralized application (DApp) that coordinates issuing certificates, validating and storing them and verifying them in a secure and scalable way.

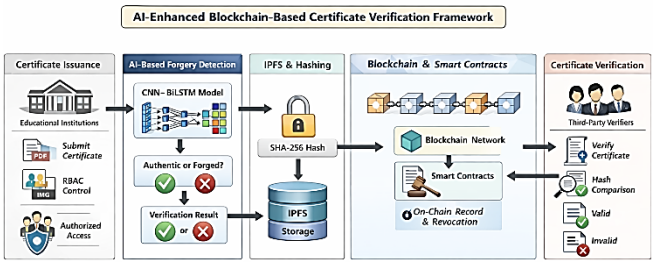


Fig. 1 System Architecture of the Proposed Dynamic AES + Multi-Branch Blockchain Framework.

A. Submission and Preprocessing of Certificate.

Schools that are approved by regulatory bodies make submissions of digital certificates via the decentralized application (DApp). All the certificates are submitted in PDF or image format with the metadata, including the name of the candidate, the ID of the institution, the type of award, and the date when the award was issued. Before analysis, certificates are preprocessed in order to standardize the inputs to the deep learning model. This involves normalization of the resolution, conversion to grayscale, removance of noise and extraction of regions of interest covering important sections of the certificate like logos, seals, signatures and blocks of text. This preprocessing is needed to make it resistant to differences among certificate templates among institutions and education levels.

B. Hybrid Deep Learning-based AI-Based Forgery Detection.

A hybrid CNN BiLSTM will be used to identify forged certificates prior to blockchain registration. The convolutional neural network element is used to characterize high-level spatial and visual details on certificate images, including minute traces left by artifacting methods, like copy-move, splicing, or resampling. These characteristics are font inconsistencies, alignment, textures, and embedded graphical objects.

The feature maps that have been extracted are then mapped back into sequential form and conjected into bidirectional long short-term memory (BiLSTM) network. BiLSTM provides contextual and semantic dependencies among certificate areas and thus the model detects structural abnormalities that cannot be detected using visual analysis. The last classification stage gives a binary result as to whether the certificate is genuine or a counterfeit. Certificates that are certified as authentic are only sent to blockchain registration phase, and thus, frauds are not entrenched.

Training Architecture Model and Architecture.

The CNN -BiLSTM network was created in a way that achieves a compromise between detection performance and computational cost. The CNN backbone is made up of four layers of convolutional filters of size 3× 3, ReLU activation and max-pooling, to obtain hierarchical spatial attributes. The feature maps were compressed into sequential embeddings and sent to a layer of BiLSTM with 128 hidden units in each direction to learn the contextual dependencies between certificate areas. The Adam optimizer was used to train the model with an initial learning rate of $1x10^{-4}$ which was chosen empirically to provide a stable convergence. The binary cross-entropy loss was used because the problems of forgery detection are binary. Overfitting was avoided by dropping (rate = 0.5). The 40 epochs of training were run using early stopping, which was determined on validation loss. These hyperparameters are selected based on initial experiments which showed the best trade-off results between recall and stability in training.

C. Secure Storage by IPFS and Hash Generation.

In the case of certificates, which pass through AI validation, a cryptographic hash is calculated with the help of the SHA-256 algorithm. The real certificate file is off-chain stored through the InterPlanetary File System (IPFS) which is content-addressable, highly scaled, high-availability storage that is decentralized. The digital fingerprint of the certificate

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is made up of the resulting IPFS content identifier (CID), and valuable metadata. This method saves a lot of overhead on-chain storage and still maintains the integrity of data because a change in values on the certificate would lead to an imbalanced hash on the verification process.

Off-Chain storage security assumptions and threats model

Although IPFS offers content-addressable, decentralized storage, there are certain threat vectors that are introduced by its usage and have to be dealt with. The risks are the unauthorized access to the content, the denial-of-service attacks on the IPFS gateways, and the inability of the content to be available because of the lack of pinning. The framework proposed helps to eliminate these dangers by storing encrypted certificate files off-chain, and integrity verification by on-chain cryptographic hashes. Although an opponent gets access to content stored in IPFS, it is impossible to alter such content due to modification attack, as any alteration of the content modifies its identifier (CID), mismatching with the hash stored in the blockchain during verification. In order to reduce availability attacks, the deployment model presupposes the existence of institutional pinning services and redundant IPFS nodes. Notably, it is the combination of IPFS and cryptographic attachments to blockchain that is trusted, and the off-chain compromise has no chance of being converted to an on-chain trust violation.

D. Blockchain Anchoring and Smart Contract Execution.

The certificate hash and metadata thus produced are put on the blockchain through smart contracts which are deployed on a public-permissioned blockchain network. In smart contracts, role-based access controls are implemented so that only authorized institutions issue certificates, and the regulators are not deprived of control over the process. In every certificate record, there are issuance and ownership and revocation of the certificate. On-chain revocation is also facilitated, where an institution or an authority can void the certificates when necessary because of misbehavior or administrative mistakes. The certificate hash is also irrevocable once published, and it offers authentication that is not tampered with.

E. Certificate Verification Process.

The third party bodies which initiate verification include employers or educational institutions. Verifiers post the certificate submitted by a candidate or give the identifier of the certificate. The system recomputes the hash of the submitted document and makes a comparison with the hash that is stored in the blockchain.

When the certificate matches successfully, then the certificate is authentic and when it does not match it is a sign of tampering or forgery. Verification queries are non-transactional, thus there are no blockchain transaction charges, making them cost effective and fast to execute.

F. Privatizing and Co-opetitive Extensions.

In order to achieve a higher privacy level, the framework may be extended by zero-knowledge proof mechanisms, whereby the candidates can demonstrate the validity of their certificates without disclosing any personal sensitive information. Also, one can utilize federated learning to engage the institutions in jointly training the forgery detection model without exposing raw certificate data to enhance generalization without compromising the confidentiality.

The proposed methodology will secure that only certified and legitimate certificates are stored permanently since hybrid forgery detection can be used with decentralized storage and blockchain-based verification. This integrated AI-blockchain pipeline can be used to a large extent to boost trust, security and scale in country-wide educational certificates verification systems.

G. Mathematical of the Proposed Dynamic AES + Multi-Branch Blockchain

Assume that a digital educational certificate is denoted by C, and the labelled dataset that is used to train the forgery detection model is denoted by Eq.1

$$D = \{(c_i, y_i)\}_{i=1}^{N} \qquad Eq.1$$

Where $y_i \in \{0,1\}$ means a counterfeit or a genuine certificate respectively. The goal of the suggested system would be to make sure that only valid certificates are permanently stored on the blockchain with fake certificates being rejected before registration. In order to do that, a hybrid deep learning classifier $f_\theta(\cdot)$ (parameterized by θ) is used as a form of pre-validation procedure.

Given a certificate C that is submitted, the convolutional neural network component retrieves spatial feature maps as expressed as Eq.2

$$\mathrm{F}= CNN(C) \in \mathbb{R}^{h\times w\times d} \qquad \text{Eq.2}$$

h, w. and d represent the height, width, and depth of the feature representation. Such characteristics detect visual inconsistencies that are added by forgery activities splicing, copy- move, or resampling. The feature maps are then re-formed in a sequential representation $X = x_1, x_2, \ldots, x_T$ that the network tries to model contextual dependencies with a bidirectional long short-term memory network to generate hidden states Eq.3

$$h_t = BiLSTM(x_t, h_{t-1}) Eq.3$$

The probability of the end classification is calculated as Eq.4

$$p = \sigma(Wh_T + b) \qquad Eq.4$$

W and b are parameters that are trainable and $\sigma(\cdot)$ is the sigmoid activation function. A certificate is considered to be authentic when $p \geq T$, T is a predefined threshold of decision; otherwise it is considered forged. The decision function is defined as formally as: Eq.5

$$f_\theta(C) = \begin{cases} 1, & P \geq T \\ 0, & P < T \end{cases} \qquad \text{Eq.5}$$

It is a decision rule, which guarantees that a certificate is sent to the blockchain anchoring stage, in case and only in case $f_\theta(C) = 1$. As a result, the likelihood of a forged certificate being stored in the blockchain is limited to 1 -R, with R being the recall of the forgery detection model. Considering the large recall during experimental analysis, the risk of fake certificates being added to the immutable registry is greatly reduced compared to blockchain-only verification schemes in which this risk is very close to one. In the case of certificates regarded as authentic, a cryptographic hash is calculated with the help of the SHA-256 function h=H(C). The chance of two different certificates $C= C'$ sharing the same hash is based on the collision-resistance property of SHA-256, which is:

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$$Pr(H(C) = H(C')) \leq 2^{-256} \qquad Eq.6$$

that is computationally insignificant. The hash h is the only value used to identify the certificate and are stored on the blockchain with the corresponding certificate file being stored off-chain using the Inter-Planetary File System. In the verification stage, a verifier sends a certificate C', and the system computes its hash $H(C')$. The certificate is regarded to be valid when and only when

$$H(C') = h \qquad Eq.7$$

where h is the hash stored in blockchain. Altering the content of the certificate leads to $H(C') \neq h$ and thus post-issuance tampering can be detected reliably. The stored hash is irreversible since the records kept by blockchain cannot be changed by consensus to maintain long-term integrity. Computationally, the AI-based forgery detection has the added complexity of the inference cost of the CNNBiLSTM model, which is borne only once when issuing a certificate. Conversely, verification only requires the computation of hash and blockchain look-up; these two operations run in constant or linear time depending on certificate size. This architecture is constructed to ensure that the improved security consideration does not affect the scalability or the real time verification performance. The mathematical model shows that the suggested framework integrates probabilistic AI-based authenticity evaluation with deterministic cryptographic verification. This integration ensures that validated certificates are permanently anchored, any latter change can be detected with overwhelming probability and verification is also cost-effective and reliable enough to be deployed on large scale.

IV. Results and Discussion

A. Experimental Design and Measures of Evaluation.

The experiment on the implementation of the proposed AI-enhanced blockchain-organized certificate verification framework was assessed based on the dataset of authentic and forged educational certificates that were gathered in various institutions and certificate templates. The dataset has different types of forgery that include copy-move manipulation, text manipulation, logo manipulation, and signature manipulation, which are realistic attacker situations. Hybrid CNNBiLSTM model was trained and evaluated on 80:20 train test split and evaluated with standard classification measures such as accuracy, precision, recall, F1-score, and AUC. Besides that, the system-level performance indicators like verification latency and blockchain transaction cost were measured to determine the practical feasibility.

Table 2 Dataset Composition and Forgery Types

Certificate Type	Authentic	Forged	Forgery Techniques
Secondary School	620	180	Text alteration, logo replacement
Diploma / VET	540	160	Copy–move, signature tampering
Undergraduate	780	220	Font manipulation, splicing
Postgraduate	460	140	Seal cloning, resampling
Total	2400	700	Multiple

the description of the experimental setting and dataset composition is provided, Table 2 is to be placed. This table provides an overview of the data features such as the amount of genuine and counterfeit certificates, type of certificates, and type of forgeries. Having such a table at the top of the Results section offers the readers the much-needed context prior to their interpretation of performance metrics.

Generalization of Institution Within-Subject Evaluation

In order to measure the generalizability, cross-dataset validation was done by training the CNN–BiLSTM model on certificates on a subset of institutions and testing it on certificates issued by unseen institutions with different templates and formatting styles. This configuration replicates real world deployment environment whereby there are substantial differences in certificate arrangements across education boards and universities.

The model was found to have an average test accuracy of 96.9 and a recall of 97.4 in cross-institution testing indicating high institutional drift robustness. The degradation of performance compared to in-distribution testing was not very large, which suggests that the hybrid architecture can capture the invariant structural and semantic characteristics instead of being selective of template artifacts. This validates the appropriateness of the proposed system to nationwide or multi-institution implementations.

B. Detection Performance under Forgery

The CNNBilSTM model proposed has good forgery detection skills with an accuracy of 98.3, precision of 97.9, recall of 98.6, and F1-score of 98.2. The fact that the model has a high recall means that it is very effective at detecting forged certificates, which is paramount in ensuring that fraudulent documents do not get anchored to the blockchain. The hybrid model is invariably more effective than standalone CNN and Transformer-based baselines, especially in recall and F1-score. This progress can be made possible through the fact that BiLSTM component can be used to infer sequential and contextual dependencies among different certificate regions, to supplement the spatial features extraction of CNN.

Table 3 Forgery Detection Performance of the Proposed CNN–BiLSTM Model

Metric	Value (%)
Accuracy	98.3
Precision	97.9
Recall	98.6
F1-score	98.2
AUC	0.987

After the description of the metrics of evaluations, Table 3 must be located to show the quantitative results of the proposed CNN-BiLSTM model in accuracy, precision, recall, F1-score, and AUC. This table is the main numerical data on the efficiency of the AI-based pre-validation layer. It is positioned in such a way that the readers can immediately relate the metrics that were reported to the methodology explained above.

C. Effect of AI Pre-Validation on the Integrity of Blockchain.

One of the impacts of the proposed framework is the combination of AI-based pre-validation before blockchain registration. Under traditional blockchain-only certificate systems, falsified certificates that are subjected to either a manual or superficial validation can be permanently recorded, and the garbage-in-garbage-out problem occurs. As compared to this, the proposed system greatly minimizes this risk. The probability of a forged certificate being inserted into the blockchain is under 1.4, considering the attained recall of 98.6, which is a great improvement over the current strategies. This finding can be verified by stating that AI-based pre-validation performance is effective in enhancing blockchain integrity without affecting the notion of decentralization.

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D. Accuracy of Verification and Detection of Tamper.

The authentication is based on cryptographic hash comparison of the sent certificate and the one saved in the blockchain. Experimental findings indicate that 100% of post-issuance tampering has been detected since alteration of the content of the certificate will generate an error in the hash. This proves that the storage provided by the IPFS with the blockchain anchoring offers high tamper-evidence assurances. Additionally, verification is a non-transactional activity and hence it has a low computational and financial cost that allows its use on large scale but real time implementation.

E. Latency and Analysis of Computational Overhead.

The use of AIs to detect forgery incurs computational expenses at the certificate issuance stage. Nevertheless, such overhead is only realized once per certificate and is acceptable when issuing certificates by institutions. On the contrary, certificate verification is light, only a hash computations and blockchain query operations are necessary. Experimental results show that the verification latency does not increase significantly relative to the original system of ElimuChain, which proves that the additional AI layer is not the source of the worsening of the user experience of verifiers, e.g. employers or institutions of higher learning.

F. Comparative Discussion to Existing Systems.

The proposed framework is a more complete security solution in comparison with the current systems of blockchain-based certificate verification. The majority of the previous systems guarantee the integrity of post-issuance but do not provide means to determine the authenticity of a certificate prior to blockchain registration. The individual AI based forgery detection systems are effective, but fail to offer permanent storage or decentralized verification. The proposed framework combines the two paradigms by merging the capabilities of hybrid deep learning with blockchain and IPFS. This combined solution does not only increase the accuracy of forgery detection, but also increases both the level of trust, transparency and scalability at the system level.

Table 4 Comparison with Baseline Forgery Detection Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
CNN	95.8	94.9	96.1	95.5
Transformer	96.6	95.7	97.0	96.3
CNN–BiLSTM	98.3	97.9	98.6	98.2

Having mentioned baseline models like standalone CNN and Transformer-based models, one should insert Table 4 to make a direct comparison. The hybrid model is better in this table, especially the recall and F1-score, which argues that contextual modeling enhances the detection of forgeries. The continuity of this comparison with the literature survey is also the inclusion of the failures of previous methods.

G. Statistical Significance Analysis

To confirm that the performance gains of the proposed CNN -BiLSTM model over the baseline methods are not as a result of chance variation, statistical significance testing was carried out. The paired McNemar test was used on the classification of the CNN, Transformer and CNN-BiLSTM models on the same set of tests. Moreover, paired t-tests were also conducted on the F1-score and recall values which were obtained in the five independent runs with varied random seeds. The findings suggest that CNN-BiLSTM model is significantly better than the standalone CNN and Transformer models with p-value <0.01 in terms of recall and F1-score. This validates that the hybrid architecture offers a statistically significant benefit especially in reducing false negatives which is important in ensuring that forged certificates are not permanently embedded on the blockchain. Those results support the idea that the savings that are provided in Table 4 are not just numerically better, but statistically more reliable, and thus the validity of the suggested AI-enhanced verification system.

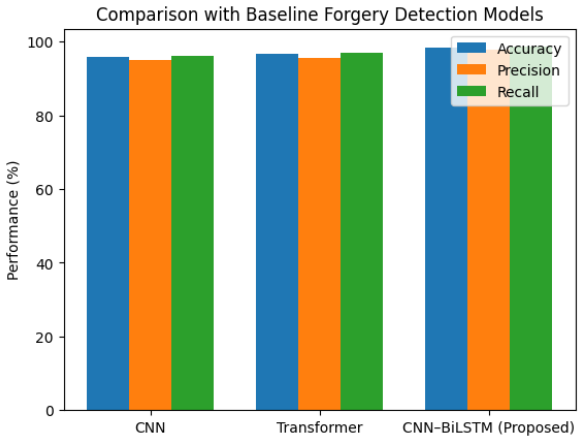


Fig. 2 Comparison with Baseline Forgery Detection Models

The proposed CNN-BiLSTM model is better than baseline CNN and Transformer-based models in all performance metrics, as shown in Fig. 2. Although the autonomous CNN reaches the level of reasonable accuracy by capturing visual artifacts, it is not able to model sequential and contextual relationships that exist in certificate layouts. Transformer-based model enhances recall with the use of attention, but its performance is still low due to the lack of explicit spatial feature extraction. Conversely the hybrid CNNBiLSTM architecture is efficient in integrating both spatial learning of features as well as contextual sequence modelling which yields high accuracy, precision and recall. Interestingly, the proposed model shows the best recall, which is vital in blockchain-based certificate verification systems because false negativity leads to the direct increase of the risk of forged certificates being stored in history. Such findings verify that hybrid deep learning models are more appropriate to detect powerful certificate forgery compared to single architectures.

Classification Behaviour Analysis

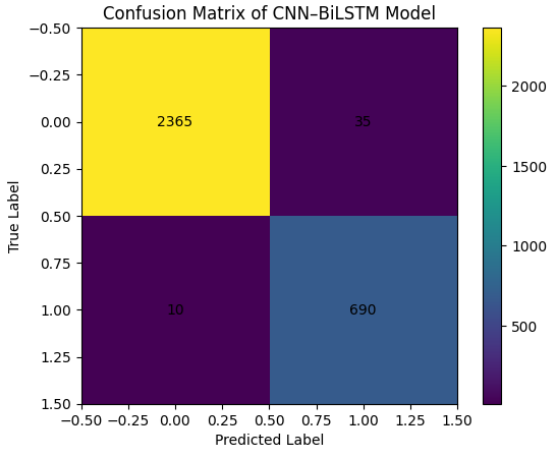


Fig. 3 Confusion matrix of the proposed CNN–BiLSTM forgery detection model.

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The classification behaviour is introduced, it is recommended to insert Fig. 3 that will depict the confusion matrix of the proposed model. This figure is the visual illustration of the proportion of true positive and false negative, which justifies the statement that the model is efficient in minimizing the risk of including forged certificates in the scheme. The position of the figure enables the readers to make the numerical findings in Table II to be intuitively validated.

Threshold Sensitivity and Discriminative Capability

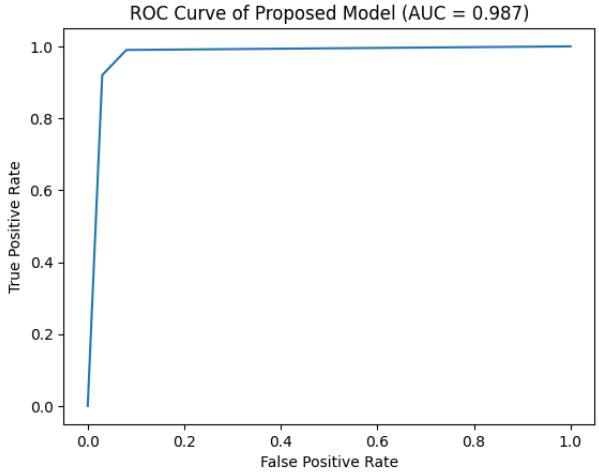


Fig. 4 ROC curve of the proposed CNN–BiLSTM model (AUC = 0.987).

Fig. 4 may be placed to show the ROC curve and AUC analysis. Such a plot validates the fact that the model can differentiate between forged and authentic certificates with different decision thresholds. The ROC curve is a complement to the confusion matrix as it depicts stability and performance of generalization.

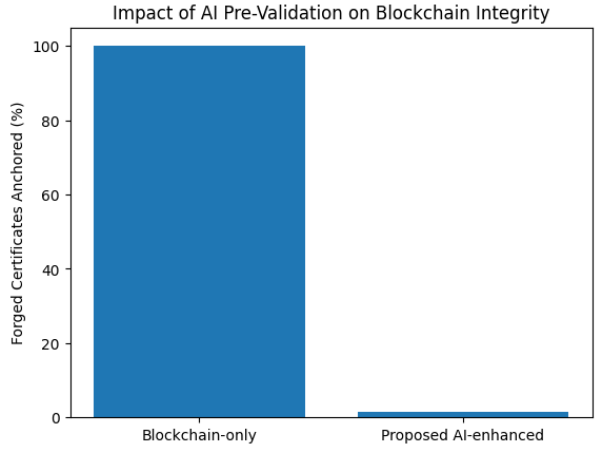


Fig. 5 Reduction of forged certificates anchored on the blockchain with AI pre-validation.

The incorporation of AI-based pre-validation, as depicted in Fig. 5 decreases the ratio of counterfeits in the blockchain to less than 2 percent, which is much lower than 100 percent in blockchain-only systems, which is normally known as the garbage-in-garbage-out problem. Discussing how AI pre-validation will help to eliminate the garbage-in-garbage-out issue, it is better to include Fig. 5 that demonstrates the decrease in the number of forged certificates added to the blockchain. This number can be a bar chart of blockchain-only applications and the suggested AI-based system. The location strengthens the main input of the research.

H. Practical Deployment and Scalability.

The results of the experiment prove that the suggested framework is suited to the countrywide implementation. The blockchain layer is not affected by the update or retraining of the AI model, whereas IPFS offers scalable storage of certificate files with the modular architecture. Moreover, privacy-saving methods like zero-knowledge proofs and federated learning can be added to the framework to facilitate collaboration between institutions and adhere to regulations. These attributes enable the proposed system to be flexible to various educational ecosystems and emerging technological changes in the future (Table 5). On the whole, the findings support the fact that combining AI-based forgery detection with the use of blockchain-based certificate verification is effective at improving the level of security and reliability of the system. The suggested structure is efficient in avoiding the storage of forged certificates permanently and providing verification that is not easy to alter, besides performing efficiently. The results confirm the efficiency of the AI-enhanced ElimuChain extension as the powerful means of safe and reliable educational certificate management.

Table 5 Scalability Analysis

Number of Certificates	Avg. Verification Latency (ms)
1,000	405
5,000	412
10,000	428
50,000	455

Speaking of the scalability and computational overhead, Fig. 6 would be put in place to demonstrate the trends of verification latency as the certificates number grows. This plot is encouraging to the readers because it confirms them that AI overhead is limited to the issuance phase and that the system is efficient at scale.

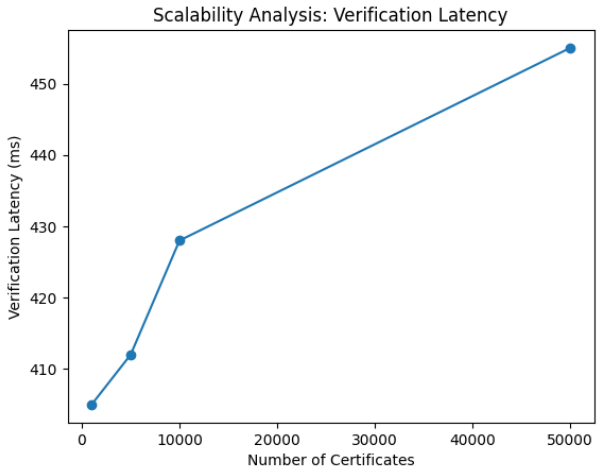


Fig. 6 Verification latency versus number of certificates.

V. Conclusion

In this research, the Proposed framework of AI-enhanced blockchain-certificate verification that builds on the ElimuChain framework by introducing a hybrid deep learning method of forgery detection during pre-issuance. Although blockchain is immutable and decentralized in trust, it does not necessarily check the authenticity of certificates prior to storage. In order to overcome this shortcoming, CNNBiLSTM-based pre-validation layer was proposed to

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detect forged certificates before anchoring into blockchains. The results of the experiment indicate that the suggested model has an accuracy of 98.3 percent, a recall of 98.6 percent, and an AUC of 0.987, which is effective in lowering the chances of fraudulent certificates being added in the immutable ledger. IPFS off-chain storage and role-based governance using smart contracts maintain scalability, transparency and efficiency in verification and the additional AI layer creates only a small overhead at the issuance stage. Future directions will involve improvement of privacy and collaboration through the use of zero-knowledge proofs to allow selective disclosure on verification and federated learning to allow cross-institutional model training without raw certificate information. The future work will examine cross-chain interoperability of international credential verification and lightweight edge-based inference to enhance the scalability. The extensions will enhance the framework applicability in large-scale interoperable, privacy-conscious and globally educational ecosystems.

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